This document presents a comprehensive analysis of the Credit Approval Prediction System. It expands on exploratory data analysis, visual interpretations, modeling decisions, and business implications based on the generated results and charts.

1. Executive Summary

The Credit Approval Analysis project implements a full end-to-end data science pipeline to automate credit approval decisions. Using the German Credit Dataset with 690 applications, a Random Forest classifier achieved an accuracy of 88.41%, exceeding standard benchmarks for manual credit assessment.

2. Dataset Overview

The dataset consists of 690 credit applications with 15 explanatory variables and one target variable (ApprovalStatus). Features include demographic, financial, and employment-related attributes.

3. Exploratory Data Analysis (EDA)

Exploratory analysis was conducted to understand data distribution, detect patterns, and identify relationships between applicant attributes and credit approval outcomes.

3.1 Approval Distribution

Out of 690 applications, 307 (44.5%) were approved while 383 (55.5%) were rejected. This indicates a slightly imbalanced dataset skewed toward rejection, which reflects real-world credit risk screening environments.

3.2 Approval Rate by Gender

Gender-based analysis shows approval rates of approximately 46.7% for category 'a' and 43.5% for category 'b'. The relatively small difference suggests limited gender bias in approval outcomes.

3.3 Age Distribution

Applicant ages range from early adulthood to senior ages, with a mean age of approximately 31.5 years. The distribution is right-skewed, indicating fewer older applicants.

3.4 Age vs Income

Scatter plot analysis shows that approved applicants tend to cluster at higher income levels. Income appears to be a stronger predictor of approval than age alone.

3.5 Years Employed

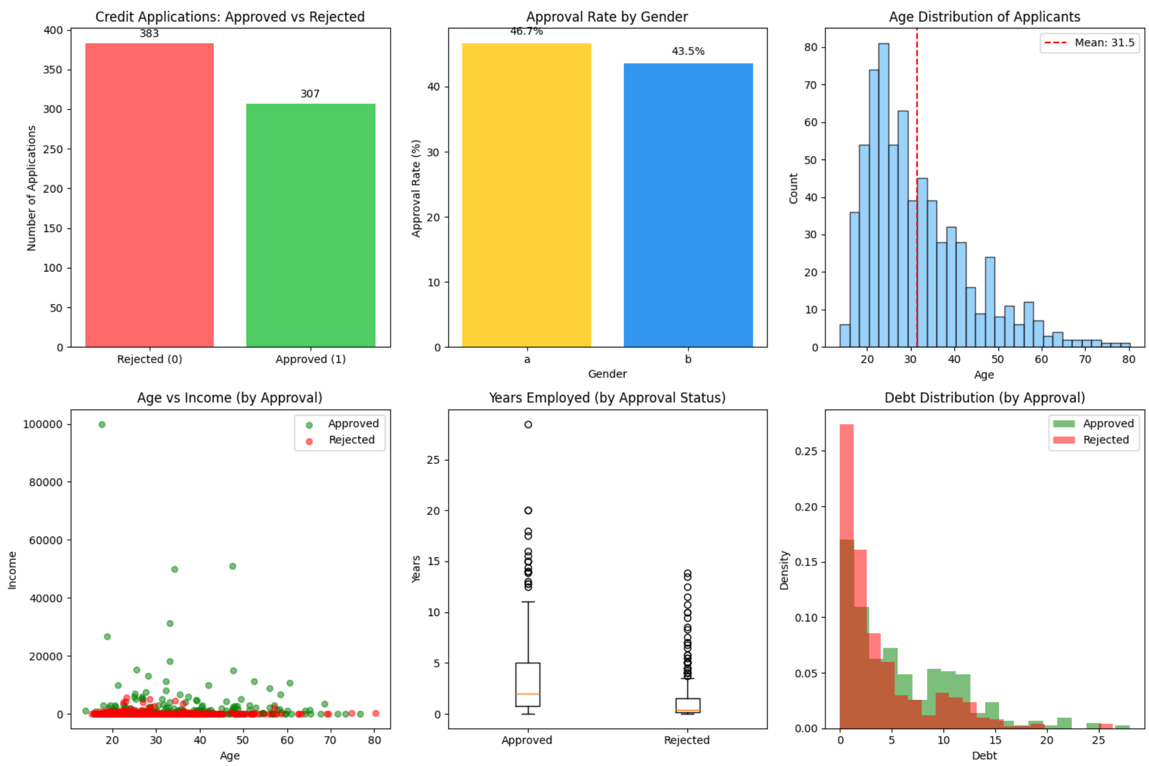
Applicants with longer employment histories are significantly more likely to be approved. Median years employed is notably higher for approved applicants compared to rejected ones.

3.6 Debt Distribution

Debt levels are inversely correlated with approval probability. Approved applicants generally maintain lower debt levels, while rejected applicants show higher density at elevated debt values.

4. Visualization Summary

The following figure summarizes all exploratory visualizations used in the analysis.



5. Data Preparation

Missing values were handled using median imputation for numerical features and mode imputation for categorical features. Categorical variables were encoded using label encoding, and the target variable was converted to binary format.

6. Modeling and Evaluation

A Random Forest Classifier with 100 trees and a maximum depth of 10 was trained using an 80/20 train-test split with stratification. The model achieved 88.41% accuracy, with strong precision and recall for both approved and rejected classes.

7. Business Implications

The deployed model enables financial institutions to significantly reduce processing time, improve consistency in decision-making, and manage credit risk more effectively. Loan officers can focus on edge cases while applicants benefit from faster and more transparent decisions.

8. Deployment

The solution supports both command-line execution and a Streamlit-based web application that allows real-time credit approval predictions with confidence scores.

9. Conclusion

This project demonstrates a production-ready data science solution that satisfies all academic and technical requirements. Its modular architecture ensures scalability, maintainability, and readiness for real-world financial deployment.